Small-time Controllability of Control Systems on 3D Lie Groups

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Abstract

We study affine control systems with one control on 3D unimodular Lie groups. Such control systems are first classified into a finite number of equivalent systems. After completing this classification, we determine the small-time controllability of each of the equivalent systems.

1 Introduction

In the article [2], Agrachev, Kazandjian, and Pozzoli considered the Schrödinger equation for a quantum harmonic oscillator. Based on this equation, the paper studied the controllability of left-invariant control affine systems of the form

$$\frac{d}{dt}q(t) = q(t)\Big(a + u_0(t)b + \sum_{i=1}^{d} u_i(t)X_i + r(t)Z\Big) \tag{1.1}$$

where $q \in SL_2 \ltimes H_d$. Notabaly, the article proved that the system (1.1) is small-time controllable on $SL_2 \ltimes H_d$, and that (1.1) with $u_i, r = 0$ for $i \in \{1, \dots, d\}$ is small-time controllable on SL_2. The small-time controllability of (1.1) played an important role in the construction of small-time controllable examples of the bilinear Schrödinger equation.

This result motivates the question: does there exist 3D unimodular Lie groups other than $SL(2)$ for which the result holds true?

To answer this question, we study affine control systems with one control on 3D unimodular Lie groups, taking the form

$$\dot{q}(t) = (A + u(t)B)q(t) \tag{1.2}$$

with $q \in G$, $A, B \in \mathfrak{g}$, and u the control. G is a 3D unimodular Lie group, which is completely classified into the Lie groups $SU(2)$, $SL(2)$, H, $SE(2)$, and $SH(2)$. $\mathfrak{g}$ is the corresponding Lie algebra. See Appendix A for definitions of the 3D unimodular Lie groups, their corresponding Lie algebras, the basis elements of the Lie algebras, and the commutator relations.

Section 2 is concerned with classification, where we classify system (1.2) on each of these Lie groups into normal forms, or equivalent systems, of which there are only finitely many. Section 3 begins by formally defines small-time controllability, among other concepts from geometric control theory. Then, we determine the small-time controllability of each of the normal forms discoverd. Through this exhaustive search, we find that the only small-time controllable examples occur in $SL(2)$.

Theorem 1.1. Consider the control system (1.1) with $u_i, r = 0$ for $i \in \{1, \dots, d\}$. It is completely classified into 26 normal forms, listed in Table 1 in Appendix A. In particular, $(E - F, H)$ and (F, E) in $SL(2)$ are the only normal forms of (1.1) that are small-time controllable.

2 Classification of Control Systems

Consider the control system (1.2) with $q \in G$ and $A, B \in \mathfrak{g}$, as formulated in the introduction. When classifying (1.2) on a Lie group, we want to simplify the pair (A, B) into an equivalent pair, say $(\tilde{A}, \tilde{B})$, such that the original system is small-time controllable if and only if the equivalent system is small-time controllable. This equivalency is denoted $(A, B) \sim (\tilde{A}, \tilde{B})$.

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The classification process makes use of the natural symmetries of the problem. The first are automorphisms of Lie algebras $\Phi : \mathfrak{g} \to \mathfrak{g}$, leading to $(A, B) \sim (\Phi(A), \Phi(B))$. In particular, inner automorphisms, which act by conjugation by some $g \in G$, give $(A, B) \sim (gAg^{-1}, gBg^{-1})$ when applied simultaneously on A and B. In some cases, we will also utilise outer automorphisms. Second, the control can be shifted by $\delta \in \mathbb{R}$, so

$$A + uB = A + (u' + \delta)B = (A + \delta B) + u'B$$

The control can further be scaled by a constant $\mu \in \mathbb{R}$, giving

$$A + uB = A + (\mu u')B = A + u'(\mu B)$$

Finally, the time can be scaled by a factor of $\lambda \in \mathbb{R}^*$ so that $(A, B) \sim (\lambda A, \lambda B)$. This last equivalence is not immediately obvious. To see this, first consider the $\lambda > 0$ case. Consider the trajectory $q(t)$ that is the solution to (1.2), and another trajectory $x(t) = q(\lambda t)$ resulting from scaling time by a factor of λ. Then

$$\dot{x}(t) = \frac{d}{dt} q(\lambda t) = \lambda \dot{q}(\lambda t) = (\lambda A + u(\lambda t)\lambda B)q(\lambda t) = (\lambda A + v(t)\lambda B)x(t)$$

Let the attainable set of (A, B) from q_0 from time $\tau \geq 0$ be $\mathcal{A}(\leq \tau, q_0)$. Let the attainable set of $(\tilde{A}, \tilde{B})$ from q_0 from time $T \geq 0$ be $\tilde{\mathcal{A}}(\leq T, q_0)$, where $T = \lambda\tau$. If (A, B) is small-time controllable, then $\mathcal{A}(\leq \tau, q_0) = G$ for every $q_0 \in G$ and $\tau > 0$. By choosing appropriate τ, we see that this is true if and only if $\tilde{\mathcal{A}}(\leq \tau, q_0) = G$ for every $q_0 \in G$ and $T > 0$. So (A, B) is small-time controllable if and only if $(\lambda A, \lambda B)$ is small-time controllable. For $\lambda < 0$ we build on the last result, and just need to further show that $(A, B) \sim (-A, -B)$. Considering the reversed trajectory $x(t) = q(T - t)$ and following the same strategy as before extends the equivalence to all $\lambda \in \mathbb{R}^*$. The equivalence relations are summarised in the following definition.

Definition 2.1. For the control system (1.2), we have the following equivalence relations.

1. Automorphisms of Lie algebras: $(A, B) \sim (\Phi(A), \Phi(B))$
2. Control shifting: $(A, B) \sim ((A + \delta B), B)$ for $\delta \in \mathbb{R}$
3. Control scaling: $(A, B) \sim (A, \mu B)$ for $\mu \in \mathbb{R}$
4. Time scaling: $(A, B) \sim (\lambda A, \lambda B)$ for $\lambda \in \mathbb{R}^*$

Assumption. We will assume that B is nonzero, otherwise the control cannot act on the system. We will also assume that A and B are independent, otherwise the system can only be driven in a single direction. It is useful to combine points 3 and 4 of Definition 2.1 to obtain $(\mu A, \lambda B) \sim (A, B)$ whenever $\mu, \lambda \neq 0$, which is always true given this assumption.

To complete the proof of the classification, we will take advantage of algebraic characteristics that are invariant under the elementary transformations listed in Definition 2.1 to show that the normal forms derived are, in fact, not equivalent to each other.

Similar work regarding the classification of control systems has already been partially done by Remsing, for example in [5, 6, 4], notabaly on the $SU(2)$, $SL(2)$, H, and $SE(2)$ groups. In any case, we will provide, along with its proof, a more complete classification in this paper.

Proposition 2.2. Control systems of the form (1.2) on each 3D unimodular Lie group are equivalent to one of the normal forms for the 3D unimodular Lie group, as listed in Table 1 of Appendix A.

Proof. The demonstration for this proposition is given throughout the following sections. □

2.1 Classification of Control Systems on $SU(2)$

We consider the system (1.2) with $q \in SU(2)$ and $A, B \in \mathfrak{su}(2)$. Begin by writing A and B as linear combinations of the basis elements

$$A = a_1^0\sigma_1 + a_2^0\sigma_2 + a_3^0\sigma_3,\ B = b_1^0\sigma_1 + b_2^0\sigma_2 + b_3^0\sigma_3.$$

Let $K(\cdot, \cdot)$ denote the Killing form. $\mathfrak{su}(2)$ endowed with the Killing form is isomorphic to $\mathbb{R}^3$ with the standard Euclidean scalar product, where the isomorphism is given by $\sigma_i \mapsto e_i$, $i \in \{1, 2, 3\}$. Inner automorphisms act as rotations in $\mathbb{R}^3$. Alternatively, we can see this by noting that $\mathrm{Aut}(\mathfrak{su}(2)) = SO(3)$.

So, we construct a 3D space with σ_1, σ_2, σ_3 as the axes, analogous to $\mathbb{R}^3$. Applying a combination of conjugation group actions to both A and B, we rotate B onto the σ_3 axis, then rotate A onto the σ_1-σ_3 plane.

$$\begin{aligned}(A, B) &\sim (gAg^{-1}, gBg^{-1}) \\ &= (a_1^1\sigma_1 + a_3^1\sigma_3, b_3^1\sigma_3) \\ &\sim (a_1^1\sigma_1, b_3^1\sigma_3) \\ &\sim (a_1^1\sigma_1, a_1^1\sigma_3) \\ &\sim (\sigma_1, \sigma_3)\end{aligned}$$

Hence, all control systems (1.2) on $SU(2)$ are equivalent to (σ_1, σ_3).

2.2 Classification of Control Systems on $SL(2)$

We consider the system (1.2) with $g \in SL(2, \mathbb{R})$ and $A, B \in \mathfrak{sl}(2)$. Recall that for matrices $u_i, u_j \in \mathfrak{sl}(2)$, the matrix of the Killing form is the matrix K whose entries are $k_{ij} = K(u_i, u_j)$. Using the basis elements given in A.5, we compute the Killing form matrix to be

$$K = \begin{pmatrix} 0 & 4 & 0 \\ 4 & 0 & 0 \\ 0 & 0 & 8 \end{pmatrix}$$

The eigenvalues of this matrix are -4, 4, and 8. There are 2 positive eigenvalues and 1 negative eigenvalue, implying a metric signature $(2, 1)$, corresponding to the Lorentzian space $\mathbb{R}^{2,1}$. Only inner automorphisms will be required in this procedure, and they are characterised by

$$\mathrm{Inn}(\mathfrak{sl}(2)) = PSL(2, \mathbb{R}) \simeq SO(1, 2)$$

The inner automorphisms give all the Lorentz transformations. This is discussed in [11].

Here, the Killing form is analogous to the Minkowski scalar product. Vectors in Lorentzian space can be classified as elliptic, hyperbolic, or parabolic based on the sign of their Killing form. Let $X \in \mathfrak{sl}(2)$. If $K(X, X) = 0$, then X is parabolic, lying on the light cone. If $K(X, X) < 0$, then X is elliptic, lying on the hyperboloid of two sheets inside the light cone. If $K(X, X) > 0$, then X is hyperbolic, lying on the hyperboloid of one sheet outside the light cone.

Lorentz transformations include null rotations, rotations, and Lorentz boosts. There does not exist a Lorentz transformation between the hyperboloid of two sheets and the hyperboloid of one sheet. There also does not exist a Lorentz transformation between the upper and lower sheets of the hyperboloid of two sheets.

The strategy is to identify a representative for each class. We choose E, H, $E-F$ as the representatives for the parabolic, hyperbolic, and elliptic cases, respectively. B could be in any of these classes, giving rise to three cases to be analysed. Since this representative and B lie in the same region of space, we can apply Lorentz transformations via conjugation to simplify B to this representative. We may then proceed to simplify A.

Remark. If, in our convention, E is the representative for the upper half of the light cone, then $-E$ is the representative for the lower half of the light cone. Similarly for the hyperboloid of two sheets, we take $E - F$ and $F - E$. However, it is not necessary to discuss the upper and lower halves separately, as one can be transformed to the other by scaling and inverting time.

Begin by writing A and B as linear combinations of the basis elements

$$A = a_1^0E + a_2^0F + a_3^0H, \; B = b_1^0E + b_2^0F + b_3^0H.$$

Case 1: $K(B, B) > 0$, B is hyperbolic

By a Lorentz transformation, B can be simplified to a multiple of H.

$$\begin{aligned}(A, B) &\sim (gAg^{-1}, gBg^{-1}) \\ &= (a_1^1E + a_2^1F + a_3^1H, b_3^1H)\end{aligned}$$

If $a_1^1 \neq 0$ and $a_2^1 = 0$, then

$$\begin{aligned}(A, B) &= (a_1^1 E + a_3^1 H, b_3^1 H) \\ &\sim (a_1^1 E, b_3^1 H) \\ &\sim (a_1^1 E, a_1^1 H) \\ &\sim (E, H)\end{aligned}$$

If $a_1^1 = 0$ and $a_2^1 \neq 0$, then

$$\begin{aligned}(A, B) &\sim (a_1^1 E + a_2^1 F + a_3^1 H, b_3^1 H) \\ &= (a_2^1 F + a_3^1 H, b_3^1 H) \\ &\sim (a_2^1 F, b_3^1 H) \\ &\sim (a_2^1 F, a_2^1 H) \\ &\sim (F, H)\end{aligned}$$

If $a_1^1 \neq 0$ and $a_2^1 \neq 0$, then we need to find elements in the stabiliser group of H to continue simplifying while preserving H. Such elements of $\mathrm{Stab}(H)$ must be of the form

$$g \in \left\{ \begin{pmatrix} a & 0 \\ 0 & d \end{pmatrix} : ad = 1 \right\}$$

which represents a Lorentz boost. Applying to $(a_1^1 E + a_2^1 F + a_3^1 H, b_3^1 H)$, we have

$$\begin{aligned}(A, B) &\sim (a_1^1 E + a_2^1 F + a_3^1 H, b_3^1 H) \\ &\sim (a_1^1 a^2 E + a_2^1 d^2 F + a_3^1 H, b_3^1 H) \\ &\sim (a_1^1 a^2 E + a_2^1 d^2 F, b_3^1 H)\end{aligned}$$

If $\mathrm{sgn}(a_1^1) = \mathrm{sgn}(a_2^1)$, then, using $ad = 1$, it is possible to choose $a, d \in \mathbb{R}$ such that $a_1^1 a^2 = a_2^1 d^2 = c$, with $c \in \mathbb{R}$ being a constant with the appropriate sign. It follows that

$$\begin{aligned}(A, B) &\sim (c(E + F), b_3^1 H) \\ &\sim (c(E + F), cH) \\ &\sim (E + F, H)\end{aligned}$$

If $a_1^1 > 0$, $a_2^1 < 0$, then we can choose $a, d \in \mathbb{R}$ such that $a_1^1 a^2 = -a_2^1 d^2 = c$, with $c > 0$. Then

$$\begin{aligned}(A, B) &\sim (c(E - F), b_3^1 H) \\ &\sim (c(E - F), cH) \\ &\sim (E - F, H)\end{aligned}$$

In the case $a_1^1 < 0$ and $a_2^1 > 0$, following the same procedure gives $(-E + F, H)$, which is equivalent to $(E - F, H)$.

Case 2: $K(B, B) < 0$, B is elliptic

By a Lorentz transformation, B can be simplified to a multiple of $E - F$.

$$\begin{aligned}(A, B) &\sim (gAg^{-1}, gBg^{-1}) \\ &= (a_1^1 E + a_2^1 F + a_3^1 H, b_1^1 (E - F))\end{aligned}$$

To continue, we need elements of the stabiliser group $\mathrm{Stab}(E - F)$. These elements are

$$g \in \left\{ \begin{pmatrix} a & b \\ -b & a \end{pmatrix} : a^2 + b^2 = 1 \right\}$$

which is a rotation matrix, and so the Lorentz transformations in this case act on elements of $\mathfrak{sl}(2)$ as rotations in a plane. If we fix $E - F$ as one axis, then for a 3D space, we need to construct two more, and it is on the plane formed by these two axes that we will rotate. Consider

$$E + F = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \; H = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}$$

By computing the Killing forms, one can check that $E-F$, $E+F$, and H are indeed orthogonal, forming a set of coordinate axes. Finding these two remaining axes can be thought of as finding the orthogonal complement of B, $B^{\perp}$, spanned by $E+F$ and H. Now we rewrite A and B in terms of this new set of coordinates, reduce A onto $B^{\perp}$ by control shifting, then rotate on $B^{\perp}$. That is,

$$\begin{aligned}(A,B) &\sim (a_1^1E + a_2^1F + a_3^1H, b_1^1(E-F)) \\ &\sim (a_1^2(E+F) + a_3^2H, b_1^1(E-F)) \\ &\sim (a_1^3(E+F), b_1^1(E-F)) \\ &\sim (a_1^3(E+F), a_1^3(E-F)) \\ &\sim (2a_1^3E, a_1^3(E-F)) \\ &\sim (2a_1^3E, 2a_1^3(E-F)) \\ &\sim (E, E-F)\end{aligned}$$

Remark. We could have equivalently rotated to H, which would give $(H, E-F)$. It does not matter which one we rotate to. Indeed, we always have $K(A,A) > 0$, hence rotations are all occuring on the hyperboloid of one sheet.

Case 3: $K(B,B) = 0$, B is parabolic

By a Lorentz transformation, B can be reduced to a multiple of E.

$$\begin{aligned}(A,B) &\sim (gAg^{-1}, gBg^{-1}) \\ &= (a_1^1E + a_2^1F + a_3^1H, b_1^1E)\end{aligned}$$

Again, we find elements of the stabiliser group $\mathrm{Stab}(E)$. Such elements are of the form

$$g \in \left\{ \begin{pmatrix} a & b \\ 0 & a \end{pmatrix} : a^2 = 1 \right\}$$

representing the null rotation. Applying this to $(a_1^1E + a_2^1F + a_3^1H, b_1^1E)$,

$$(A,B) \sim ((a_1^1 - b^2a_2^1 - 2aba_3^1)E + a_2^1F + (aba_2^1 + a_3^1)H, b_1^1E)$$

If $a_2^1 = 0$, then

$$\begin{aligned}(A,B) &\sim (a_3^1H, b_1^1E) \\ &\sim (a_3^1H, a_3^1E) \\ &\sim (H, E)\end{aligned}$$

If $a_2^1 \neq 0$, then choose $b = -a_3^1/aa_2^1$ to eliminate the H component in A, giving

$$\begin{aligned}(A,B) &\sim (a_2^1F, b_1^1E) \\ &\sim (a_2^1F, a_2^1E) \\ &\sim (F, E)\end{aligned}$$

It remains to prove that the normal forms found above are not equivalent to each other. This requires the following general result on the invariance of the Killing form under both inner and outer automorphisms. See [13] for the statement and the proof.

Lemma 2.3. Let $\phi : \mathfrak{g} \to \mathfrak{g}$ be a Lie algebra automorphism, belonging to $Aut(\mathfrak{g})$. For any $X, Y \in \mathfrak{g}$, $K(\phi(X), \phi(Y)) = K(X,Y)$.

Assume (A,B) is in one of the discovered normal forms. First, we examine B, where the invariant is the sign of the Killing form $K(B,B)$. Indeed, $K(\phi(B),\phi(B)) = K(B,B)$ for all Lie algebra automorphisms $\phi \in \mathrm{Aut}(\mathfrak{sl}(2))$, and $K(\mu B, \mu B) = \mu^2 K(B,B)$ for all $\mu \in \mathbb{R}$. Since $K(H,H) > 0$, $K(E-F, E-F) < 0$, and $K(E,E) = 0$, this invariant separates the normal forms into three categories:

1. (E,H), (F,H), $(E+F,H)$, $(E-F,H)$

2. $(E, E-F)$

3. (H, E), (F, E)

For the first category, consider elements $g \in \mathrm{Stab}(H)$, whose general form has already been found. Then, explicitly computing the conjugations on E, F, $E+F$, and $E-F$ shows that it is impossible to map one to the other, proving that these four normal forms are are non-equivalent. Knowing that the Killing form must be preserved under Lie algebra automorphisms, the two normal forms in the third category are easier to distinguish, as we note that $K(H,H) > 0$, whereas $K(F,F) = 0$.

2.3 Classification of Control Systems on H

Consider the system (1.2) with $q \in H$ and $A, B \in \mathfrak{h}$. We start by writing A and B as linear combinations of the basis elements

$$A = a_1^0 X + a_2^0 Y + a_3^0 Z,\ B = b_1^0 X + b_2^0 Y + b_3^0 Z$$

The Heisenberg group is different from the previous two scenarios in that the Killing form is degenerate. Firstly, we may use inner automorphisms. These preserve the X and Y components, only shifting the Z component. Taking

$$g = \begin{pmatrix} 1 & a & c \\ 0 & 1 & b \\ 0 & 0 & 1 \end{pmatrix}$$

and applying this transformation to A and B, we obtain

$$gAg^{-1} = a_1^0 X + a_2^0 Y + (-ba_1^0 + aa_2^0 + a_3^0)Z,\ gBg^{-1} = b_1^0 X + b_2^0 Y + (-bb_1^0 + ab_2^0 + b_3^0)Z$$

The second type of transformations are symplectic maps, which, in fact, belong to the outer automorphisms. The symplectic map uses a matrix $T \in Sp(2,\mathbb{R})$ to act by left matrix multiplication on the (X,Y) vector in A and B, and preserves the Z component. Symbolically, we can write the mapping $((X,Y),Z) \mapsto (T(X,Y),Z)$. Note that in the $n=2$ case, $Sp(2,\mathbb{R}) = SL(2,\mathbb{R})$.

Consider also the centre of the Lie algebra $\mathfrak{h}$, which is $\mathfrak{z}(\mathfrak{h}) = \{cZ : c \in \mathbb{R}\}$. We try to simplify B first, and to do so, we analyse two cases: either $B \in \mathfrak{z}(\mathfrak{h})$, or $B \notin \mathfrak{z}(\mathfrak{h})$.

Case 1: $B \in \mathfrak{z}(\mathfrak{h})$

Here, $b_3^0 \neq 0$, while $b_1^0 = b_2^0 = 0$. Employ a symplectic matrix

$$T = \begin{pmatrix} a & b \\ c & d \end{pmatrix},\ ad - bc = 1$$

where $c = -a_2^0$, $d = a_1^0$, and a, b are chosen to satisfy $ad - bc = 1$. This symplectic map transforms the X and Y components of A into X. Therefore,

$$\begin{aligned}(A,B) &= (a_1^0 X + a_2^0 Y + a_3^0 Z, b_3^0 Z) \\ &\sim (X + a_3^0 Z, b_3^0 Z) \\ &\sim (X, b_3^0 Z) \\ &\sim (X, Z)\end{aligned}$$

Case 2: $B \notin \mathfrak{z}(\mathfrak{h})$

Here, $(b_1^0, b_2^0) \neq (0,0)$. b_3^0 may or may not be 0. In the inner automorphism on B,

$$gBg^{-1} = b_1^0 X + b_2^0 Y + (-bb_1^0 + ab_2^0 + b_3^0)Z$$

Since $(b_1^0, b_2^0) \neq (0,0)$, we can choose $a, b \in \mathbb{R}$ of the matrix g such that $-bb_1^0 + ab_2^0 + b_3^0 = 0$, so

$$gBg^{-1} = b_1^0 X + b_2^0 Y$$

This can further be rotated on to X by applying the symplectic map to A and B. That is,

$$\begin{aligned}(A,B) &\sim (gAg^{-1}, gBg^{-1}) \\ &= (a_1^0 X + a_2^0 Y + (-ba_1^0 + aa_2^0 + a_3^0)Z, b_1^0 X + b_2^0 Y) \\ &\sim (a_1^1 X + a_2^1 Y + (-ba_1^0 + aa_2^0 + a_3^0)Z, b_1^1 X) \\ &\sim (a_2^1 Y + (-ba_1^0 + aa_2^0 + a_3^0)Z, b_1^1 X)\end{aligned}$$

If $a_2^1 = 0$, then

$$\begin{aligned}(A,B) &= ((-ba_1^0 + aa_2^0 + a_3^0)Z, b_1^1 X) \\ &= ((-ba_1^0 + aa_2^0 + a_3^0)Z, (-ba_1^0 + aa_2^0 + a_3^0)X) \\ &= (Z, X)\end{aligned}$$

If $a_2^1 \neq 0$, then we take another $h \in H$ that is also an element of the stabiliser of X. This is

$$h = \begin{pmatrix} 1 & a' & c' \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \in \mathrm{Stab}(X)$$

We now apply the inner automorphism.

$$\begin{aligned}(A,B) &\sim (a_2^1 Y + (-ba_1^0 + aa_2^0 + a_3^0)Z, b_1^1 X) \\ &\sim (h(a_2^1 Y + (-ba_1^0 + aa_2^0 + a_3^0)Z)h^{-1}, h(b_1^1 X)h^{-1}) \\ &= (a_2^1 Y + (a_2^1 a' + (-ba_1^0 + aa_2^0 + a_3^0))Z, b_1^1 X)\end{aligned}$$

Since $a_2^1 \neq 0$, we can choose the entries of h to be $a' = -(-ba_1^0 + aa_2^0 + a_3^0)/a_2^1$, $c' \in \mathbb{R}$ so that $a_2^1 a' + (-ba_1^0 + aa_2^0 + a_3^0) = 0$. Proceeding with this,

$$\begin{aligned}(A,B) &\sim (a_2^1 Y, b_1^1 X) \\ &\sim (a_2^1 Y, a_2^1 X) \\ &\sim (Y, X)\end{aligned}$$

Now, we need to show that the three normal forms found above are non-equivalent. Starting with B, an invariant is the property $B \in \mathfrak{z}(\mathfrak{h})$, since inner and outer automorphisms both preserve the centre, and scaling also preserves the centre. So (X, Z) is not equivalent to (Z, X) or (Y, X), two cases have been separated:

1. (X, Z)
2. (Z, X), (Y, X)

It remains to distinguish between these latter two cases. The property of whether or not $A \in \mathrm{span}\{X, Y\}$ is an invariant. Indeed, inner and outer automorphisms, scaling by a constant, and adding B, which has zero Z coordinate in both cases, all preserve this property. So (Z, X) and (Y, X) are not equivalent. Therefore, all three normal forms found are non-equivalent.

2.4 Classification of Control Systems on $SE(2)$

We consider the system (1.2) with $q \in SE(2)$ and $A, B \in \mathfrak{se}(2)$. Begin by writing A and B as linear combinations of the basis elements

$$A = a_1^0 X + a_2^0 Y + a_3^0 Z,\ B = b_1^0 X + b_2^0 Y + b_3^0 Z$$

For a matrix $g \in SE(2)$, the conjugation group action on the basis elements gives

$$gXg^{-1} = \cos\theta X + \sin\theta Y,\ gYg^{-1} = -\sin\theta X + \cos\theta Y,\ gZg^{-1} = yX - xY + Z$$

with $x, y, \theta \in \mathbb{R}$. So on A and B, we have

$$gAg^{-1} = (a_1^0 \cos\theta - a_2^0 \sin\theta + a_3^0 y)X + (a_1^0 \sin\theta + a_2^0 \cos\theta - a_3^0 x)Y + a_3^0 Z$$

$$gBg^{-1} = (b_1^0 \cos\theta - b_2^0 \sin\theta + b_3^0 y)X + (b_1^0 \sin\theta + b_2^0 \cos\theta - b_3^0 x)Y + b_3^0 Z$$

There are two cases to consider: when $B \in \text{span}\{X, Y\}$, and when $B \notin \text{span}\{X, Y\}$.

Case 1: $B \in \text{span}\{X, Y\}$

In this case, $B = b_1^0 X + b_2^0 Y$, with $(b_1^0, b_2^0) \neq 0$ and $b_3^0 = 0$. Since $b_3^0 = 0$, we can disregard x and y, and choose θ so that gBg^{-1} rotates B in the X-Y plane onto X, then simplify.

$$\begin{aligned}
(A, B) &\sim (gAg^{-1}, gBg^{-1}) \\
&= ((a_1^0 \cos\theta - a_2^0 \sin\theta + a_3^0 y)X + (a_1^0 \sin\theta + a_2^0 \cos\theta - a_3^0 x)Y + a_3^0 Z, b_1^1 X) \\
&= (a_1^1 X + a_2^1 Y + a_3^0 Z, b_1^1 X) \\
&\sim (a_2^1 Y + a_3^0 Z, b_1^1 X)
\end{aligned}$$

If $a_3^0 \neq 0$, then we require elements of $\text{Stab}(X)$, so that conjugation preserves $b_1^1 X$. It is clear that we need $\theta = 0$. Moreover, noting the result of gZg^{-1}, we choose $x = a_3^1/a_3^0$ and $y = 0$ to eliminate the Y component and prevent X from reappearing in A. Thus,

$$\begin{aligned}
(A, B) &\sim (a_3^0 Z, b_1^1 X) \\
&\sim (a_3^0 Z, a_3^0 X) \\
&\sim (Z, X)
\end{aligned}$$

If $a_3^0 = 0$, then assuming $a_2^1 \neq 0$ (otherwise we obtain the same result as before),

$$\begin{aligned}
(A, B) &\sim (a_2^1 Y, b_1^1 X) \\
&\sim (a_2^1 Y, a_2^1 X) \\
&\sim (Y, X)
\end{aligned}$$

Case 2: $B \notin \text{span}\{X, Y\}$

Here, $b_3^0 \neq 0$. When applying inner automorphisms, we first choose $x = 0$ and $y = 0$ so that we only rotate in the X-Y plane.

$$\begin{aligned}
(A, B) &\sim (gAg^{-1}, gBg^{-1}) \\
&= ((a_1^0 \cos\theta - a_2^0 \sin\theta + a_3^0 y)X + (a_1^0 \sin\theta + a_2^0 \cos\theta - a_3^0 x)Y + a_3^0 Z, b_1^1 X + b_3^0 Z) \\
&= (a_1^1 X + a_2^1 Y + a_3^0 Z, b_1^1 X + b_3^0 Z)
\end{aligned}$$

If $b_1^1 = 0$, then

$$\begin{aligned}
(A, B) &\sim (a_1^1 X + a_2^1 Y + a_3^0 Z, b_3^0 Z) \\
&\sim (a_1^1 X + a_2^1 Y, b_3^0 Z)
\end{aligned}$$

Apply a second inner automorphism, choosing $x = 0$, $y = 0$, and $\theta \in \mathbb{R}$ to rotate onto X.

$$\begin{aligned}
(A, B) &\sim (a_1^2 X, b_3^0 Z) \\
&\sim (a_1^2 X, a_1^2 Z) \\
&\sim (X, Z)
\end{aligned}$$

If $b_1^1 \neq 0$, choose another $g \in SE(2)$ with $\theta = 0$ so that the inner automorphism causes no rotation, only translation. Choosing $x = 0$, $y = -b_1^1/b_3^0$,

$$\begin{aligned}
(A, B) &\sim (g(a_1^1 X + a_2^1 Y + a_3^0 Z)g^{-1}, g(b_1^1 X + b_3^0 Z)g^{-1}) \\
&= ((a_1^1 + a_3^0 y)X + (a_2^1 - a_3^0 x)Y + a_3^0 Z, (b_1^1 + b_3^0 y)X - b_3^0 xY + b_3^0 Z) \\
&\sim (a_1^2 X + a_2^1 Y + a_3^0 Z, b_3^0 Z) \\
&\sim (a_1^2 X + a_2^1 Y, b_3^0 Z)
\end{aligned}$$

Finally, use one last inner automorphism and set $x = 0$, $y = 0$. $\theta \in \mathbb{R}$ is chosen such that $a_1^2 X + a_2^1 Y$ is rotated on to X.

$$\begin{aligned}
(A, B) &\sim (a_1^3 X, b_3^0 Z) \\
&\sim (a_1^3 X, a_1^3 Z) \\
&\sim (X, Z)
\end{aligned}$$

Lastly, we prove that the above normal forms are not equivalent to each other. Beginning with B, an invariant is the property that $B \in [\mathfrak{se}(2), \mathfrak{se}(2)] = \text{span}\{X, Y\}$. This separates

1. (Y, X), (Z, X)
2. (X, Z)

To distinguish between (Y, X) and (Z, X), consider A, for which the invariant property we identify is the same as in the Heisenberg group case. Indeed, whether the Z direction vanishes is preserved under automorphisms, scaling, and adding B, which does not contain Z in either of the two normal forms in question. From this invariant, it is clear that there does not exist a transformation that maps (Y, X) to (Z, X). This concludes the proof.

2.5 Classification of Control Systems on $SH(2)$

Consider the system (1.2) with $q \in SH(2)$ and $A, B \in \mathfrak{sh}(2)$. The $SH(2)$ group is similar to $SE(2)$, except that $SH(2)$ is translations and hyperbolic rotations preserving the Lorentzian.

Again, begin by writing A and B as linear combinations of the basis elements

$$A = a_1^0 X + a_2^0 Y + a_3^0 Z,\ B = b_1^0 X + b_2^0 Y + b_3^0 Z$$

For a matrix $g \in SH(2)$, the conjugation group action on the basis elements gives

$$gXg^{-1} = \cosh\theta X + \sinh\theta Y,\ gYg^{-1} = \sinh\theta X + \cosh\theta Y,\ gZg^{-1} = -yX - xY + Z$$

with $x, y, \theta \in \mathbb{R}$. Indeed, these transformations are combinations of hyperbolic rotations determined by θ, and translations determined by x and y. For A and B, we have

$$gAg^{-1} = (a_1^0 \cosh\theta + a_2^0 \sinh\theta - a_3^0 y)X + (a_1^0 \sinh\theta + a_2^0 \cosh\theta - a_3^0 x)Y + a_3^0 Z$$

$$gBg^{-1} = (b_1^0 \cosh\theta + b_2^0 \sinh\theta - b_3^0 y)X + (b_1^0 \sinh\theta + b_2^0 \cosh\theta - b_3^0 x)Y + b_3^0 Z$$

The analysis requires two cases to be considered: $B \in \text{span}\{X, Y\}$ and $B \notin \text{span}\{X, Y\}$.

Case 1: $B \in \text{span}\{X, Y\}$

Here, $b_3^0 = 0$ and $(b_1^0, b_2^0) \neq (0, 0)$. We would like to begin by applying hyperbolic rotations to simplify B. Since $b_3^0 = 0$, to rotate B onto X or Y, we need to solve the equation $\tanh\theta = -b_2^0/b_1^0$ or $\tanh\theta = -b_1^0/b_2^0$. tanh is a mapping $\mathbb{R} \to (-1, 1)$, so at most one of these two equations can be solved, depending on the relative magnitudes $|b_1^0|$ and $|b_2^0|$. Three different cases are to be analysed.

If $|b_1^0| = |b_2^0|$, then none of the equations have a solution, and it is not possible to rotate B onto only X or only Y. We proceed as follows. If $a_3^0 = 0$, then

$$\begin{aligned}(A, B) &= (a_1^0 X + a_2^0 Y, b_1^0 X + b_2^0 Y) \\ &\sim (a_1^1 X, b_1^0 X + b_2^0 Y) \\ &= (a_1^1 X, b_1^0 (X \pm Y)) \\ &\sim (a_1^1 X, a_1^1 (X \pm Y)) \\ &\sim (X, X \pm Y)\end{aligned}$$

If $a_3^0 \neq 0$, apply a translation using a $g \in SH(2)$ with $\theta = 0$, $x, y \in \mathbb{R}$.

$$\begin{aligned}(A, B) &= (a_1^0 X + a_2^0 Y + a_3^0 Z, b_1^0 X + b_2^0 Y) \\ &\sim ((a_1^0 - a_3^0 y)X + (a_2^0 - a_3^0 x)Y + a_3^0 Z, b_1^0 (X \pm Y)) \\ &= (a_3^0 Z, b_1^0 (X \pm Y)) \\ &\sim (a_3^0 Z, a_3^0 (X \pm Y)) \\ &\sim (Z, X \pm Y)\end{aligned}$$

We were able to translate A to only Z precisely because $a_3^0 \neq 0$, which allowed choices of y and x such that the coefficients of X and Y vanish.

If $|b_1^0| < |b_2^0|$, then $b_1^0 \cosh\theta + b_2^0 \sinh\theta = 0$ has a solution, and B can be rotated onto Y. If $a_3^0 = 0$, then

$$\begin{aligned}(A,B) &= (a_1^0 X + a_2^0 Y, b_1^0 X + b_2^0 Y) \\ &\sim (a_1^1 X + a_2^1 Y, b_2^1 Y) \\ &\sim (a_1^1 X, b_2^1 Y) \\ &\sim (a_1^1 X, a_1^1 Y) \\ &\sim (X, Y)\end{aligned}$$

Otherwise, if $a_3^0 \neq 0$, then

$$\begin{aligned}(A,B) &= (a_1^0 X + a_2^0 Y + a_3^0 Z, b_1^0 X + b_2^0 Y) \\ &\sim (a_1^1 X + a_2^1 Y + a_3^0 Z, b_2^1 Y) \\ &\sim (a_1^1 X + a_3^0 Z, b_2^1 Y) \\ &\sim (a_3^0 Z, b_2^1 Y) \\ &\sim (a_3^0 Z, a_3^0 Y) \\ &\sim (Z, Y)\end{aligned}$$

where the fourth step is a translation by choosing $\theta = 0$, $x = 0$, $y = a_1^1/a_3^0$.

If $|b_1^0| > |b_2^0|$, then $b_1^0 \sinh\theta + b_2^0 \cosh\theta = 0$ has a solution, and B can be rotated onto X. If $a_3^0 = 0$, then

$$\begin{aligned}(A,B) &= (a_1^0 X + a_2^0 Y, b_1^0 X + b_2^0 Y) \\ &\sim (a_1^1 X + a_2^1 Y, b_1^1 X) \\ &\sim (a_2^1 Y, b_1^1 X) \\ &\sim (a_2^1 Y, a_2^1 X) \\ &\sim (Y, X)\end{aligned}$$

Otherwise, if $a_3^0 \neq 0$, then

$$\begin{aligned}(A,B) &= (a_1^0 X + a_2^0 Y + a_3^0 Z, b_1^0 X + b_2^0 Y) \\ &\sim (a_1^1 X + a_2^1 Y + a_3^0 Z, b_1^1 X) \\ &\sim (a_2^1 Y + a_3^0 Z, b_1^1 X) \\ &\sim (a_3^0 Z, b_1^1 X) \\ &\sim (a_3^0 Z, a_3^0 X) \\ &\sim (Z, X)\end{aligned}$$

where the fourth step is a translation by choosing $\theta = 0$, $x = a_2^1/a_3^0$, $y = 0$.

Case 2: $B \notin \operatorname{span}\{X, Y\}$

Here, $b_3^0 \neq 0$. Recalling the general form of gAg^{-1} and gBg^{-1} that was initially derived, it is possible to choose $x, y \in \mathbb{R}$ such that the coefficients $b_1^0 \cosh\theta + b_2^0 \sinh\theta - b_3^0 y = 0$ and $b_1^0 \sinh\theta + b_2^0 \cosh\theta - b_3^0 x = 0$, causing the X and Y components in B to vanish. We have

$$\begin{aligned}(A,B) &\sim (gAg^{-1}, gBg^{-1}) \\ &= (a_1^1 X + a_2^1 Y + a_3^0 Z, b_3^0 Z) \\ &\sim (a_1^1 X + a_2^1 Y, b_3^0 Z)\end{aligned}$$

We may assume $(a_1^1, a_2^1) \neq (0,0)$. Now, hyperbolic rotations will be applied with the intent of reducing A, instead of B. Regardless, the arguments from Case 1 still apply, and we aim to solve $\tanh\theta = -a_1^0/a_2^0$ or $\tanh\theta = -a_2^0/a_1^0$. We analyse three different cases based on the relative magnitudes $|a_1^1|$ and $|a_2^1|$.

If $|a_1^1| = |a_2^1|$, then neither of the two equations has a solution. So we proceed as

$$\begin{aligned}(A,B) &\sim (a_1^1(X \pm Y), b_3^0 Z) \\ &\sim (a_1^1(X \pm Y), a_1^1 Z) \\ &\sim (X \pm Y, Z)\end{aligned}$$

If $|a_1^1| < |a_2^1|$, then $a_1^1 \cosh\theta + a_2^1 \sinh\theta = 0$ has a solution and we can rotate onto Y. Note that we must choose $x = y = 0$ during the transformation so that $b_3^0 Z$ is preserved.

$$\begin{aligned}(A, B) &\sim (a_2^2 Y, b_3^0 Z)\\ &\sim (a_2^2 Y, a_2^2 Z)\\ &\sim (Y, Z)\end{aligned}$$

If $|a_1^1| > |a_2^1|$, then $a_1^1 \sinh\theta + a_2^1 \cosh\theta = 0$ has a solution and we can rotate onto X. Again choose $x = y = 0$ so that the $b_3^0 Z$ is preserved.

$$\begin{aligned}(A, B) &\sim (a_1^2 X, b_3^0 Z)\\ &\sim (a_1^2 X, a_1^2 Z)\\ &\sim (X, Z)\end{aligned}$$

Finally, we show that the above normal forms are non-equivalent. Considering B first, an invariant is the property whether B belongs to the derived algebra $[\mathfrak{sh}(2), \mathfrak{sh}(2)] = \text{span}\{X, Y\}$. This separates the normal forms into two categories:

1. (X, Z), (Y, Z), $(X + Y, Z)$, $(X - Y, Z)$
2. $(X, X + Y)$, $(X, X - Y)$, $(Z, X + Y)$, $(Z, X - Y)$, (Z, Y), (X, Y), (Z, X), (Y, X)

We examine A next. In the first case, elements $g \in \text{Stab}(Z)$ are of the form

$$g = \begin{pmatrix} \cosh\theta & \sinh\theta & 0 \\ \sinh\theta & \cosh\theta & 0 \\ 0 & 0 & 1 \end{pmatrix}, \quad \theta \in \mathbb{R}$$

These are Lorentz boosts, and clearly, conjugation by g cannot map X, Y, $X+Y$, or $X-Y$ to each other. Thus, the normal forms in the first category are non-equivalent. In fact, for any $x, y \in \mathbb{R}$, the conjugation gXg^{-1}, gYg^{-1}, $g(X + Y)g^{-1}$, $g(X - Y)g^{-1}$ yield the same result as the case where $x = y = 0$. So the second category can be further split into four categories which are not equivalent to each other, namely when B equals X, Y, $X + Y$, or $X - Y$. Moreover, each category contains two members, and since the Killing form is preserved under conjugation, each of these pairs are non-equivalent. This proves that all the normal forms discovered are indeed non-equivalent.

3 Small-Time Controllability

Consider control systems of the form

$$\dot{q}(t) = (A + u(t)B)q(t), \quad q \in G, u(t) \in \mathbb{R} \tag{3.1}$$

where $A, B \in \mathfrak{g}$ are one of the normal forms listed in Table 1. For each of the normal forms found, we wish to determine its small-time controllability. Let us first recall some definitions, as given in [7].

Definition 3.1. Reachable (attainable) sets starting from q_0:

- the reachable set from q_0 at time $\tau \geq 0$ is

$$\mathcal{A}(\tau, q_0) = \{q_1 \in \mathbb{R}^n : \exists u(\cdot) : [0, \tau] \to U, q(\tau, q_0, u) = q_1\};$$

- the reachable set from q_0 within time $\tau \geq 0$ is

$$\mathcal{A}(\leq \tau, q_0) = \cup_{t \in [0,\tau]} \mathcal{A}(t, q_0);$$

- the reachable set from q_0 is

$$\mathcal{A}(q_0) = \cup_{t \in [0,\infty)} \mathcal{A}(t, q_0).$$

Definition 3.2. The system (3.1) is said to be

- controllable if for every $q_0 \in G$, $\mathcal{A}(q_0) = G$;

- small-time controllable if at the identity I, for every $\tau > 0$, we have $\mathcal{A}(\leq \tau, I) = G$;
- small-time locally controllable at q_0 if q_0 belongs to the interior of $\mathcal{A}(\leq \tau, q_0)$ for every $\tau > 0$.

Now, we state the Rashevsky-Chow theorem in the context of control theory, as formulated in [12]. It uses the bracket-generating condition to characterise the small-time controllability of a system. See also [1] and [8] for this theorem.

Theorem 3.3 (Rashevsky-Chow)**.** Let M be a smooth manifold and $f_1, \ldots, f_m$ be m smooth, i.e. C^∞ vector fields on M. Assume that

$$\text{Lie}\{f_1, \ldots, f_m\}(q) = T_q M, \quad \forall q \in M$$

Then the control system

$$\dot{q} = \sum_{i=1}^{m} u_i f_i(q)$$

is locally controllable in arbitrary small time at every point of M.

Remark. If M is connected, then local controllability implies global controllability. A proof is given in [12]. This is useful as all the matrix Lie groups considered in this paper are connected.

Small-time controllability will be studied at the identity, that is, taking $q(0) = \text{Id}$. This is always possible by considering the translation mapping $q \mapsto qh$, where $h \in G$. This mapping translates the original trajectory to another trajectory with the same control.

The goal is to apply the Rashevsky-Chow theorem to conclude small-time controllability. However, the problem is that the theorem applies only to driftless systems, while the system (3.1) clearly has a drift term A. Broadly speaking, the strategy to resolve this problem is to employ fast-oscillating controls and study the remaining drift. This will primarily be based on [3, 9].

First, we must check the bracket-generating condition. In the 3D case, if (A, B) satisfies

$$\text{span}\{A, B, [A, B]\} = \mathfrak{g}$$

then (A, B) is said to be bracket generating.

Lemma 3.4. If the pair (A, B) is not bracket generating, then the system (3.1) is not controllable, and in particular, not small-time controllable.

Proof. We use the Frobenius theorem, which states that a distribution is integrable if and only if it is involutive. The theorem and its proof can be found in sources such as [1] and [8].

If (A, B) is not bracket generating, i.e. $[A, B] \in \text{span}\{A, B\}$, then the distribution Δ is involutive. By the Frobenius theorem, there exists an integral manifold N_q through every point $q \in G$. In particular,

$$\dim(N_q) = \text{rank}(\Delta) = 2$$

while $\dim(\mathfrak{g}) = 3$. Starting on a point in such an integral manifold, one can never leave it, so the system is not controllable, and in particular, not small-time controllable. □

On the other hand, if the bracket-generating condition is satisfied, then we may proceed to the next step. A vector in $\mathfrak{g}$ is considered to be a "fast direction" if it can be made arbitrarily large by applying an arbitrarily large control. For example, B in system (3.1) is a fast direction since we are free to choose any control $u(t) \in \mathbb{R}$.

Let us first study a general result. In [3], Agrachev considered the control affine system

$$\dot{q}(t) = f_0(q(t)) + \sum_{i=1}^{k} u_i(t) f_i(q(t)), \quad q \in M, u_i \in \mathbb{R} \tag{3.2}$$

where M is a connected manifold, and $f_j \in M$, $j = 0, \ldots, k$ are smooth vector fields on M. Define the Lie algebra generated by the fast directions, that is the vector fields $f_1, \ldots, f_k$, by

$$L = \text{Lie}\{f_1, \ldots, f_k\}$$

Define $\mathcal{L} \subset \operatorname{Diff} M$ to be the Lie group of diffeomorphisms of M generated by L. [3] proved that the system

$$\dot{q}(t) = U(q(t)), \quad U \in \operatorname{conv}(\operatorname{Ad}_{\mathcal{L}} f_0) + L \tag{3.3}$$

is compatible with system (3.2), i.e. the reachable set of (3.3) is contained within $\bar{\mathcal{A}}$, the closure of the reachable set of the original system (3.2).

In the context of this article, we are studying the system (3.1). We begin by defining $L = \operatorname{Lie}\{B\} = \mathbb{R}B$, and $\mathcal{L} \subset \operatorname{Diff} G$. Then the system

$$\dot{q}(t) = U(q(t)), \quad U \in \operatorname{conv}(\operatorname{Ad}_{\mathcal{L}} A) + L \tag{3.4}$$

is compatible with (3.1). Here, the adjoint is computed as $\operatorname{Ad}_{\mathcal{L}} A = \{e^{sB} A e^{-sB}\}$. Now, we follow the iterative procedure outlined in [3], where we iteratively compute the convexified adjoint orbit and search for new fast directions to exploit, in hopes of obtaining a driftless system that is compatible with the original system, in which case we can apply the Rashevsky-Chow theorem. We first quotient the fast directions from $\operatorname{conv}(\operatorname{Ad}_{\mathcal{L}} A) + L$, resulting in the convexified adjoint orbit $\operatorname{conv}(\operatorname{Ad}_{\mathcal{L}} A)$. We search for new directions in this orbit, characterised by an infinite line through the origin. If this new fast direction and the original fast direction B generates the third direction needed to span the Lie algebra $\mathfrak{g}$, then we are done. Otherwise, we repeat the previous process. Note that since there are now two fast directions, L and $\mathcal{L}$ must be updated accordingly, and we must quotient both fast directions from $\operatorname{conv}(\operatorname{Ad}_{\mathcal{L}} A) + L$. This process is repeated at most one more time in the 3D scenario. In the ideal situation, we eventually obtain a system of the form

$$\dot{q}(t) = (u(t)A + v(t)B + w(t)C)q(t), \quad q \in G, (u(t), v(t), w(t)) \in \mathbb{R}^3$$

where A, B, C together span the Lie algebra $\mathfrak{g}$. This is an extended system that, as proven in [3, 9], is compatible with the original system (3.1). Notice there is no longer a drift term, but rather three fast directions. This puts us in a position to apply the Rashevsky-Chow theorem to conclude small-time controllability.

We will attempt to apply this iterative procedure to each of the normal forms of system (3.1) discovered to determine their small-time controllability. The following observation will be helpful for reducing repetitiveness when it comes to some frequently recurring geometries of the convexified set.

Lemma 3.5. If $\operatorname{ad}_B^2(A) = 0$, then $\operatorname{conv}(\operatorname{Ad}_{\mathcal{L}} A)$ is a straight line not passing through the origin, given by $\{A + s[B, A] : s \in \mathbb{R}\}$.

Proof. Let $\operatorname{ad}_B^2(A) = 0$, which implies $[B, [B, A]] = 0$. Thus

$$\operatorname{Ad}_{\mathcal{L}} A = e^{\operatorname{ad}_{sB}}(A) = A + [sB, A] + \frac{1}{2}[sB, [sB, A]] + \cdots = A + s[B, A]$$

Since the pair (A, B) is bracket generating, we can conclude

$$\operatorname{conv}(\operatorname{Ad}_{\mathcal{L}} A) = \{A + s[B, A] : s \in \mathbb{R}\}$$

□

What was described above was the full iterative procedure that allowed us to successfully conclude small-time controllability. In reality, there are various points at which the procedure can terminate. For instance, if the convexified adjoint orbit is bounded, we can immediately conclude that the system (3.1) is not small-time controllable.

Lemma 3.6. Let L be the set of all fast directions available. If $(\operatorname{conv}(\operatorname{Ad}_{\mathcal{L}} A) + L)/L$ is bounded, then the system (3.1) is not small-time controllable.

Proof. We can travel arbitrarily fast in L, so we must quotient out the points that can be reached instantaneously from the space G, resulting in a homogeneous space G/L. We need to show that the set $\operatorname{conv}(\operatorname{Ad}_{\mathcal{L}} A)$ induces a control system on this homogeneous space, and the control times of both systems are the same. See [10] for the reduction procedure in quantum control. □

If the set is semi-infinite (i.e. not bounded, but also not infinite in the sense that there exists an infinite line through the origin within the set), then the situation becomes more delicate, and needs to be treated on a case-by-case basis. There are cases where it is preferable to follow another route to prove non-small-time controllability. The following method, inspired by barrier functions from convex optimisation, will be useful on numerous occasions.

Lemma 3.7. Assume there exists a barrier function F such that

1. $F(q(t)) > 0$ for all trajectories $q(t)$, $t > 0$, of the system (3.1),
2. $F(I) = 0$,
3. there exists a point X close to the identity I where $F(X) < 0$.

Then the system (3.1) is not controllable, and in particular, not small-time controllable.

Proof. Let F be a function that satisfies the three conditions above. Then there exists a point in a neighbourhood of the identity that cannot be reached by any trajectory of the system in positive time. So, the attainable set from the identity $\mathcal{A}(\leq \tau, I) \neq G$ for all $\tau > 0$. Hence the system is not small-time controllable. □

There are also cases where, while the convexified set is semi-infinite and not truly infinite, we may identify a semi-infinite line, one that begins at the origin and is infinite in one direction, which belongs to the convexification at infinity. To do so, the concept of Poisson stability is required, defined in [1] as follows. Let $\operatorname{Vec} M$ denote the set of all smooth vector fields on a smooth manifold M.

Definition 3.8. Let $f \in \operatorname{Vec} M$ be a complete vector field. A point $q \in M$ is called Poisson stable for f if for any $t > 0$ and any neighbourhood $\mathcal{O}_q$ of q there exists a point $q' \in \mathcal{O}_q$ and a time $t' > t$ such that $q' \circ e^{t'f} \in \mathcal{O}_q$

Note that this concept is also referred to as recurrent vector fields in [7]. A consequence for controllability resulting from Poisson stability is the following.

Proposition 3.9. Let $\mathcal{F} \subset \operatorname{Vec} M$ be a bracket-generating system. If a vector field $f \in \mathcal{F}$ is Poisson stable, then the field $-f$ is compatible with $\mathcal{F}$.

Proof. See the proof of Proposition 8.14 in [1]. □

The implication for this article is that at the identity, a recurrent vector field returns, after some period of time, to an arbitrarily small neighbourhood of the origin. This allows the line that was initially infinite only in one direction to be approximated as being infinite in both directions. To check whether the identity Id is Poisson stable for f, we check whether the trajectory $\mathrm{Id} \circ e^{tf} = e^{tf}$ is periodic.

Proposition 3.10. The small-time controllability of control systems of the form (3.1) on each 3D unimodular Lie group is stated in Table 1 of Appendix A.

Proof. The demonstration for this result is given throughout the following subsections. □

3.1 Controllability on $SU(2)$

First, $[\sigma_1, \sigma_3] = -\frac{1}{2}\sigma_2$, so $\operatorname{span}\{\sigma_1, \sigma_3, [\sigma_1, \sigma_3]\} = \mathfrak{su}(2)$. The pair is bracket generating, so we may proceed to the next step. Define the set $L = \operatorname{Lie}\{\sigma_3\} = \mathbb{R}\sigma_3$, and define $\mathcal{L} = \{e^{s\sigma_3} q : s \in \mathbb{R}\}$. It is easy to compute

$$\operatorname{Ad}_{\mathcal{L}} \sigma_1 = e^{s\sigma_3}\sigma_1 e^{-s\sigma_3} = \{\cos(s)\sigma_1 + \sin(s)\sigma_2 : s \in \mathbb{R}\}$$

We then convexify this adjoint orbit and quotient out the fast direction σ_3, giving

$$(\operatorname{conv}(Ad_{\mathcal{L}}\sigma_1) + L)/\mathbb{R}\sigma_3 = \{(\sigma_1, \sigma_2) : \sigma_1^2 + \sigma_2^2 \leq 1\}$$

which is a disc in the σ_1-σ_2 plane bounded by the unit circle. Since the convexified adjoint orbit is bounded, by Lemma 3.6, the system (σ_1, σ_3) is not small-time controllable.

3.2 Controllability on $SL(2)$

3.2.1 (E, H)

$[E, H] = -2E \in \operatorname{span}\{E, H\}$, so by Lemma 3.4, the system is not small-time controllable.

3.2.2 (F, H)

$[F, H] = 2F \in \operatorname{span}\{F, H\}$, so by Lemma 3.4, the system is not small-time controllable.

3.2.3 $(E+F, H)$

As $[E+F, H] = -2(E-F)$, it is easy to check that this system is bracket generating. Furthermore, the convexified adjoint orbit quotient the fast direction H is the region above one branch of a hyperbola in the E-F plane, expressed as

$$\operatorname{conv}(\operatorname{Ad}_{\overline{e^{sH}}}(E+F))/\mathbb{R}H = \operatorname{conv}\{e^{2s}E + e^{-2s}F : v \in \mathbb{R}\}$$

Such a set poses various obstacles, and instead of continuing along this route, it is considerably easier to employ the barrier function method instead. Let

$$q = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in SL(2)$$

Consider the function

$$F = ac + bd$$

Now we transform the matrix representation into a state-space representation. From

$$\dot{q} = (E + F + uH)q$$

we obtain

$$\begin{cases} \dot{a} &= c + au \\ \dot{b} &= d + bu \\ \dot{c} &= a - cu \\ \dot{d} &= b - du \end{cases}$$

Using this, compute the derivative

$$\dot{F}(q(t)) = \dot{a}c + a\dot{c} + \dot{b}d + b\dot{d} = a^2 + b^2 + c^2 + d^2 \geq 0$$

In fact, given the restriction $ad - bc = 1$, it must be that $\dot{F}(q(t)) > 0$ for all trajectories $g(t)$ of the system $(E+F, H)$. We can also check that at the identity, $F(I) = 1 \cdot 0 + 0 \cdot 1 = 0$. Lastly, consider

$$\exp(-\varepsilon F) = \begin{pmatrix} 1 & 0 \\ -\varepsilon & 1 \end{pmatrix},$$

arbitrarily close to the identity. We have $F(\exp(-\varepsilon F)) = -\varepsilon < 0$ for all $\varepsilon > 0$. By Lemma 3.7, $(E+F, H)$ is not small-time controllable.

3.2.4 $(E-F, H)$

Since $[E-F, H] = -2(E+F)$, the pair $(E-F, H)$ is bracket generating. We proceed to find the convexified adjoint orbit.

$$\operatorname{conv}(\operatorname{Ad}_{\overline{e^{sH}}}(E-F))/\mathbb{R}H = \operatorname{conv}\{e^{2s}E - e^{-2s}F : s \in \mathbb{R}\}$$

This is the region below one branch of a hyperbola in the fourth quadrant of the E-F plane. We attempt to go to the infinity in the $E-F$ direction within the convex hull, but notice that this is only the positive half of an infinite line through the origin.

For sufficiently large $c > 0$, $c(E-F)$ belongs to the convexification. Moreover, by explicitly computing the matrix exponential $e^{c(E-F)t}$, we see that the result is 2π-periodic, returning periodically to the identity. This shows $E-F$ is a recurrent vector field. We already checked that $(E-F, H)$ is a bracket generating system. By Proposition 3.9, $-(E-F)$ is compatible with the system $(E-F, H)$, and so extending the half-line to an infinite line through the origin. Finally, we have obtained the extended system

$$\dot{q}(t) = (u(t)(E-F) + v(t)H + w(t)(E+F))q(t)$$

which is compatible with the original system $(E-F, H)$. By the Rashevsky-Chow theorem, $(E-F, H)$ is small-time controllable.

3.2.5 $(E, E-F)$

$[E, E-F] = -H$, so $\operatorname{span}\{E, E-F, [E, E-F]\} = \mathfrak{sl}(2)$, and so the pair is bracket generating. We can proceed to find the convexified adjoint orbit, then quotient out the fast direction $E-F$. First, we compute the adjoint and write it in a favourable form.

$$\begin{aligned}\operatorname{Ad}_{\overline{e^{s(E-F)}}} E &= \{\cos^2(s)E - \sin^2(s)F + \sin(s)\cos(s)H : s \in \mathbb{R}\} \\ &= \{(\cos^2(s) - \sin^2(s))E + \sin^2(s)E - \sin^2(s)F + \sin(s)\cos(s)H : s \in \mathbb{R}\} \\ &= \left\{ \cos(2s)E + \sin^2(s)(E-F) + \frac{1}{2}\sin(2s)H : s \in \mathbb{R} \right\}\end{aligned}$$

From this, the convexified adjoint orbit can be expressed explicitly.

$$\operatorname{conv}(\operatorname{Ad}_{\overline{e^{s(E-F)}}} E)/\mathbb{R}(E-F) = \{(E, H) : E^2 + 4H^2 \leq 1\}$$

which describes the region bounded by an ellipse in the E-H plane. Hence, by Lemma 3.6, the system is not small-time controllable.

3.2.6 (H, E)

$[H, E] = 2E$, so by Lemma 3.4, the system is not small-time controllable.

3.2.7 (F, E)

$[F, E] = -H$, so the pair is bracket generating. We proceed to find the convexified adjoint orbit and quotient out the fast direction E. This is a straight line not passing through the origin, given by

$$\operatorname{conv}(\operatorname{Ad}_{\overline{e^{vE}}} F)/\mathbb{R}E = \{F + vH : v \in \mathbb{R}\}$$

Taking the fast direction into account again as in the original system, we have

$$\operatorname{conv}(\operatorname{Ad}_{\overline{e^{vE}}} F) + uE = \{F + uE + vH : u, v \in \mathbb{R}\}$$

Going to the infinity along this convex hull, we can approximate an infinite line along H in both directions. We check the commutator $[H, E] = 2E \in \operatorname{span}\{H, E\}$. E and H were unable to generate a new direction, so we need to continue to find quotient spaces, which gives us new directions to use. Now consider a modified system

$$\dot{q}(t) = (F + uE + vH)q(t)$$

This new system has two fast directions, E and H, which we quotient from the convexified adjoint orbit. This is done by computing

$$\operatorname{conv}(\operatorname{Ad}_{\overline{e^{uE+vH}}} F)/\operatorname{span}\{E, H\} = \{e^{-2v}F : v \in \mathbb{R}\}$$

This one-dimensional set is half of an infinite line in the positive F direction. For $w > 0$ sufficiently large, wF belongs to the convexification. Consider also uE with $u = -w < 0$. Computing explicitly, $\exp((wF + uE)t) = \exp(w(F-E)t)$ passes through the origin and is 2π-periodic. Hence $F - E$ is a recurrent vector field. The triple $\{F, E, H\}$ is clearly bracket generating. Since $F - E \in \operatorname{span}\{F, E, H\}$, By Proposition 3.9, $E - F$ is compatible with the modified system, which is compatible with the original system. Therefore, the extended system

$$\dot{q}(t) = (w(t)F + u(t)E + v(t)H)q(t)$$

which is compatible with the original system. Applying the Rashevsky-Chow theorem, we conclude that the system (F, E) is small-time controllable.

3.3 Controllability on H

3.3.1 (X, Z)

$[X, Z] = 0$, so by Lemma 3.4, the system is not small-time controllable.

3.3.2 (Z, X)

$[Z, X] = 0$, so by Lemma 3.4, the system is not small-time controllable.

3.3.3 (Y, X)

Since $[Y, X] = -Z$, the pair is clearly bracket generating. We proceed to compute the convexified adjoint orbit, and quotient out the fast direction X. This is a straight line not passing through the origin, given by

$$\operatorname{conv}(\operatorname{Ad}_{\overline{e^{vX}}} Y)/\mathbb{R}X = \{Y + vZ : v \in \mathbb{R}\}$$

Taking the fast direction into account again as in the original system, we have

$$\operatorname{conv}(\operatorname{Ad}_{\overline{e^{vX}}} Y) + uX = \{Y + uX + vZ : u, v \in \mathbb{R}\}$$

Now, the X and Z directions can be made arbitrarily fast by taking large controls u and v. Checking the bracket $[Z, X] = 0 \in \operatorname{span}\{Z, X\}$, we see that X and Z were unable to generate a new direction. Repeating this procedure, we must continue to find quotient spaces, which makes available new directions to use. Now consider the modified system

$$\dot{g} = (Y + uX + vZ)g$$

Since X and Z can be made arbitrarily large, the drift Y is effectively negligable, making this system compatible with the original system, i.e. the closure of its reachable set is the same as the original system. This new system has two fast directions, X and Z, which we quotient from the convexified adjoint orbit. This is done by computing

$$\operatorname{conv}(\operatorname{Ad}_{\overline{e^{uX+vZ}}} Y)/\operatorname{span}\{X, Z\}$$

First, it is easy to compute

$$\operatorname{Ad}_{\overline{e^{uX+vZ}}} Y = \{Y + uZ : u \in \mathbb{R}\}$$

Its convex hull quotient the fast X and Z directions is

$$\operatorname{conv}(\operatorname{Ad}_{\overline{e^{uX+vZ}}} Y) = \{Y + uZ : u \in \mathbb{R}\}/\operatorname{span}\{X, Z\} = \{Y\}$$

This is a singleton, so by Lemma 3.6, the system is not small-time controllable.

3.4 Controllability on $SE(2)$

3.4.1 (Y, X)

$[Y, X] = 0$, so by Lemma 3.4, the system is not small-time controllable.

3.4.2 (Z, X)

$[Z, X] = Y$, thus the pair is evidently bracket generating. The convexified adjoint orbit after we quotient out the fast direction X, is

$$\operatorname{conv}(\operatorname{Ad}_{\overline{e^{vX}}} Z)/\mathbb{R}X = \{Z - vY : v \in \mathbb{R}\}$$

which is a straight line not passing through the origin. Taking the fast direction into account again, we obtain

$$\operatorname{conv}(\operatorname{Ad}_{\overline{e^{vX}}} Z) + uX = \{Z + uX - vY : u, v \in \mathbb{R}\}$$

Now, the X and Y directions can be made arbitrarily fast by taking large u and v. This effectively makes Z negligable. We check the commutator $[Y, X] = 0 \in \operatorname{span}\{Y, X\}$. X and Y were unable to generate a new direction, so we need to continue to find quotient spaces, to obtain new directions to use. Now we consider a modified system

$$\dot{g} = (Z + uX - vY)g$$

that is compatible with the original system (through the same reasoning as before, taking large X and Y to make Z negligable). This system has two fast directions, X and Y, which we quotient from the convexified adjoint orbit. That is,

$$\operatorname{conv}(Ad_{\overline{e^{uX-vY}}} Z)/\operatorname{span}\{X, Y\}$$

We first compute

$$\mathrm{Ad}_{\overline{e^{uX-vY}}} Z = \{Z - vX - uY : u, v \in \mathbb{R}\}$$

Convexifying it gives a plane passing through Z that is parallel to the X-Y plane. We then quotient out the fast X and Y directions, giving

$$\mathrm{conv}(\mathrm{Ad}_{\overline{e^{uX-vY}}} Z)/\operatorname{span}\{X, Y\} = \{Z\}$$

which is a singleton. By Lemma 3.6 the system is not small-time controllable.

3.4.3 (X, Z)

$[X, Z] = -Y$, and so $\operatorname{span}\{X, Z, [X, Z]\} = \mathfrak{se}(2)$, meaning the pair is bracket generating. We proceed to compute the convexified adjoint orbit. The adjoint orbit is

$$\mathrm{Ad}_{\overline{e^{sZ}}} X = \{\cos(s)X + \sin(s)Y : s \in \mathbb{R}\},$$

and its convex hull quotient the fast direction Z is

$$\mathrm{conv}(\mathrm{Ad}_{\overline{e^{sZ}}} X)/\mathbb{R}Z = \{(X, Y) : X^2 + Y^2 \leq 1\},$$

which describes the disc bounded by the unit circle in the X-Y plane. Therefore, by Lemma 3.6, the system is not small-time controllable.

3.5 Controllability on $SH(2)$

3.5.1 (X, Z)

$[X, Z] = -Y$, hence the system is bracket generating. One can then check that the convexified adjoint orbit, after we quotient out the fast direction Z, is the region to the right of one branch of a hyperbola in the X-Y plane, given by

$$\mathrm{conv}(\mathrm{Ad}_{\overline{e^{vZ}}} X)/\mathbb{R}Z = \mathrm{conv}\{\cosh vX + \sinh vY : v \in \mathbb{R}\}$$

Similar to the $(E + F, H)$ case in $SL(2)$, we attempt to apply the barrier function method. Consider the function

$$F = x \cosh\theta - y \sinh\theta$$

Now, let us transform the matrix representation of the system (X, Z) into the state-space representation. From

$$\dot{g} = (X + uZ)g$$

we obtain

$$\begin{cases} \dot{x} &= uy + 1 \\ \dot{y} &= ux \\ \dot{\theta} &= u \end{cases}$$

Using the above choice of F, $F(I) = 0$ at the identity,

$$\dot{F}(g(t)) = \dot{x}\cosh\theta + x\sinh\theta\dot{\theta} - \dot{y}\sinh\theta - y\cosh\theta\dot{\theta} = \cosh\theta \geq 1 > 0$$

for all trajectories $g(t)$ of the system for $t > 0$, and $F(\exp(-\varepsilon X)) = -\varepsilon < 0$ for all $\varepsilon > 0$ arbitrarily close to the identity. Hence, by Lemma 3.7, (X, Z) is not small-time controllable.

3.5.2 (Y, Z)

$[Y, Z] = -X$, so the pair is bracket generating. One can check that the convexified adjoint orbit, after we quotient out the fast direction Z, is the region above one branch of a hyperbola in the X-Y plane, given by

$$\mathrm{conv}(\mathrm{Ad}_{\overline{e^{vZ}}} Y)/\mathbb{R}Z = \mathrm{conv}\{\sinh vX + \cosh vY : v \in \mathbb{R}\}$$

Here, the barrier function method is almost identical to the (X, Z) case. First, the matrix differential equation

$$\dot{g} = (Y + uZ)g$$

is written in its state-space representation. That is,

$$\begin{cases} \dot{x} &= uy \\ \dot{y} &= ux + 1 \\ \dot{\theta} &= u \end{cases}$$

Consider the function

$$F(g(t)) = y\cosh\theta - x\sinh\theta$$

for which $F(I) = 0$ at the identity,

$$\dot{F}(g(t)) = \dot{y}\cosh\theta + y\sinh\theta\dot{\theta} - \dot{x}\sinh\theta - x\cosh\theta\dot{\theta} = \cosh\theta \geq 1 > 0$$

for all trajectories $g(t)$ of the system for $t > 0$, and $F(\exp(-\varepsilon Y)) = -\varepsilon < 0$ for all $\varepsilon > 0$ at a point arbitrarily close to the identity. By Lemma 3.7, the system (Y, Z) is not small-time controllable.

3.5.3 $(X + Y, Z)$

$[X + Y, Z] = -X - Y$, so by Lemma 3.4, the system is not small-time controllable.

3.5.4 $(X - Y, Z)$

$[X - Y, Z] = -Y + X$, so by Lemma 3.4, the system is not small-time controllable.

3.5.5 $(X, X + Y)$

$[X, X + Y] = 0$, so by Lemma 3.4, the system is not small-time controllable.

3.5.6 $(X, X - Y)$

$[X, X - Y] = 0$, so by Lemma 3.4, the system is not small-time controllable.

3.5.7 $(Z, X + Y)$

$[Z, X + Y] = Y + X = X + Y$, so by Lemma 3.4, the system is not small-time controllable.

3.5.8 $(Z, X - Y)$

$[Z, X - Y] = Y - X$, so by Lemma 3.4, the system is not small-time controllable.

3.5.9 (Z, Y)

$[Z, Y] = X$, therefore this pair is clearly bracket generating. Proceeding, the convexified adjoint orbit after we quotient out the fast direction Y is given by

$$\operatorname{conv}(\operatorname{Ad}_{\overline{e^{vY}}} Z)/\mathbb{R}Y = \{Z - vX : v \in \mathbb{R}\},$$

which is a straight line not passing through the origin. Taking the fast direction into account again as in the original system, we have

$$\operatorname{conv}(\operatorname{Ad}_{\overline{e^{vY}}} Z) + uY = \{Z + uY - vX : u, v \in \mathbb{R}\}$$

Now, the X and Y directions can be made arbitrarily large by taking large controls u and v. We check the commutator $[X, Y] = 0 \in \operatorname{span}\{X, Y\}$. Thus, X and Y were unable to generate a new direction. We must repeat the above process of finding quotient spaces, attempting to search for new directions to use. Consider the modified system

$$\dot{g} = (Z + uY - vX)g$$

As before, this system is compatible with the original system. This new system has two fast directions, X and Y, which we quotient from the convexified adjoint orbit. That is to say,

$$\operatorname{conv}(\operatorname{Ad}_{\overline{e^{uY - vX}}} Z)/\operatorname{span}\{X, Y\}$$

We first compute the adjoint orbit to be

$$\mathrm{Ad}_{\overline{e^{uY-vX}}} Z = \{Z - uX + vY : u, v \in \mathbb{R}\}$$

Its convexification is a plane passing through Z parallel to the X-Y plane. We then quotient out the fast X and Y directions, obtaining

$$\mathrm{conv}(\mathrm{Ad}_{\overline{e^{uY-vX}}} Z)/\operatorname{span}\{X, Y\} = \{Z\},$$

which is a singleton. By Lemma 3.6, the system is not small-time controllable.

3.5.10 (X, Y)

$[X, Y] = 0$, so by Lemma 3.4, the system is not small-time controllable.

3.5.11 (Z, X)

$[Z, X] = Y$, so the system is bracket generating, and we can proceed. The convexified adjoint orbit quotient the fast direction X is a straight line not passing through the origin. It is expressed as

$$\mathrm{conv}(\mathrm{Ad}_{\overline{e^{vX}}} Z)/\mathbb{R}X = \{Z - vY : v \in \mathbb{R}\}$$

Taking the fast direction into account again as in the original system, we have

$$\mathrm{conv}(\mathrm{Ad}_{\overline{e^{vX}}} Z) + uX = \{Z + uX - vY : u, v \in \mathbb{R}\}$$

Here, the X and Y directions can be made arbitrarily fast by taking the controls u and v to be large. Checking the commutator $[Y, X] = 0 \in \operatorname{span}\{Y, X\}$, we see that X and Y were unable to generate a new direction. We must repeat the process of finding quotient spaces, and search for possible new directions to use. To this end, consider the modified system

$$\dot{g} = (Z + uX - vY)g$$

which is compatible with the original system. This new system has two fast directions, X and Y, which we quotient from the convexified Ad orbit. We would like to determine

$$\mathrm{conv}(\mathrm{Ad}_{\overline{e^{uX-vY}}} Z)/\operatorname{span}\{X, Y\}$$

We compute the adjoint orbit first.

$$\mathrm{Ad}_{\overline{e^{uX-vY}}} Z = \{Z + vX - uY : u, v \in \mathbb{R}\}$$

Convexifying this set gives a plane passing through Z that is parallel to the X-Y plane. We then quotient out the fast X and Y directions, yielding

$$\mathrm{conv}(\mathrm{Ad}_{\overline{e^{uX-vY}}} Z)/\operatorname{span}\{X, Y\} = \{Z\},$$

a singleton. By Lemma 3.6, the system is not small-time controllable.

3.5.12 (Y, X)

$[Y, X] = 0$, so by Lemma 3.4, the system is not small-time controllable.

4 Conclusion

In this article, control systems of the form (1.2) were first classified on all 3D unimodular matrix Lie groups. 26 normal forms were discovered in total, of which one belonged to $SU(2)$, 7 belonged to $SL(2)$, 3 belonged to H, 3 belonged to $SE(2)$, and 12 belonged to $SH(2)$. Then, the small-time controllability of each of the normal forms was determined. The two normal forms that were small-time controllable both belonged to $SL(2)$, showing that $SL(2)$ is the only 3D unimodular Lie group on which the system (1.2) is small-time controllable.

A Appendix

- $SU(2)$:

$$SU(2) = \left\{ \begin{pmatrix} a & -\bar{b} \\ b & \bar{a} \end{pmatrix} : a, b \in \mathbb{C}, |a|^2 + |b|^2 = 1 \right\} \tag{A.1}$$

$$\mathfrak{su}(2) : \sigma_1 = \frac{1}{2}\begin{pmatrix} i & 0 \\ 0 & -i \end{pmatrix},\ \sigma_2 = \frac{1}{2}\begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix},\ \sigma_3 = \frac{1}{2}\begin{pmatrix} 0 & i \\ i & 0 \end{pmatrix} \tag{A.2}$$

$$[\sigma_1, \sigma_2] = \sigma_3,\ [\sigma_2, \sigma_3] = \sigma_1,\ [\sigma_3, \sigma_1] = \sigma_2 \tag{A.3}$$

- $SL(2)$:

$$SL(2) = \{X \in M_2(\mathbb{R}) : \det X = 1\} \tag{A.4}$$

$$\mathfrak{sl}(2) : E = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix},\ F = \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix},\ H = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \tag{A.5}$$

$$[H, E] = 2E,\ [H, F] = -2F,\ [E, F] = H \tag{A.6}$$

- H:

$$H = \left\{ \begin{pmatrix} 1 & a & c \\ 0 & 1 & b \\ 0 & 0 & 1 \end{pmatrix} : a, b, c \in \mathbb{R} \right\} \tag{A.7}$$

$$\mathfrak{h} : X = \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix},\ Y = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{pmatrix},\ Z = \begin{pmatrix} 0 & 0 & 1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix} \tag{A.8}$$

$$[X, Y] = Z,\ [X, Z] = 0,\ [Y, Z] = 0 \tag{A.9}$$

- $SE(2)$:

$$SE(2) = \left\{ \begin{pmatrix} \cos\theta & -\sin\theta & x \\ \sin\theta & \cos\theta & y \\ 0 & 0 & 1 \end{pmatrix} : x, y, \theta \in \mathbb{R} \right\} \tag{A.10}$$

$$\mathfrak{se}(2) : X = \begin{pmatrix} 0 & 0 & 1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix},\ Y = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{pmatrix},\ Z = \begin{pmatrix} 0 & -1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix} \tag{A.11}$$

$$[X, Y] = 0,\ [Y, Z] = X,\ [X, Z] = -Y \tag{A.12}$$

- $SH(2)$:

$$SH(2) = \left\{ \begin{pmatrix} \cosh\theta & \sinh\theta & x \\ \sinh\theta & \cosh\theta & y \\ 0 & 0 & 1 \end{pmatrix} : x, y, \theta \in \mathbb{R} \right\} \tag{A.13}$$

$$\mathfrak{sh}(2) : X = \begin{pmatrix} 0 & 0 & 1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix},\ Y = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{pmatrix},\ Z = \begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix} \tag{A.14}$$

$$[X, Y] = 0,\ [Y, Z] = -X,\ [X, Z] = -Y \tag{A.15}$$

Lie group	Normal Forms of (A, B)	STC
$SU(2)$	(σ_1, σ_3)	No
$SL(2)$	(E, H)	No
	(F, H)	No
	$(E + F, H)$	No
	$(E - F, H)$	Yes
	$(E, E - F)$	No
	(H, E)	No
	(F, E)	Yes
H	(X, Z)	No
	(Z, X)	No
	(Y, X)	No
$SE(2)$	(Y, X)	No
	(Z, X)	No
	(X, Z)	No
$SH(2)$	(X, Z)	No
	(Y, Z)	No
	$(X + Y, Z)$	No
	$(X - Y, Z)$	No
	$(X, X + Y)$	No
	$(X, X - Y)$	No
	$(Z, X + Y)$	No
	$(Z, X - Y)$	No
	(Z, Y)	No
	(X, Y)	No
	(Z, X)	No
	(Y, X)	No

Table 1: Summary of Normal Forms and Small-Time Controllability

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